

Title Page

GIG Credit
Smart Fintech Platform for freeancers
Semester Project Report
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Course:
Web Programming
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Group Members and Contribution

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Full stack development and deployment
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Quality Assurance testing and Reporting

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Project documentation, diagrams and Report
Work

Project Overview

GigCredit is a web based Fintech application made for freelancers. The idea behind this project is to help users manage their financial activities in one place. The system includes wallet management, transfers, deposits, withdrawals, expense tracking and budgeting features. The platform also keeps track of transactions and give reports through charts and dashboard summaries. Users can view their Transaction history and (next page)

manage spending more easily.

An admin panel is also part of the system for monitoring users, transactions and categories. Suspicious transactions rules are included in backend to detect unusual activities and improve security.

This project was developed using React.js for frontend, Node.js with Express.js for backend and MongoDB Atlas for database management.

Problem statement

Freelancers usually use different apps or manual methods to manage income, expenses and transactions. This makes financial tracking difficult and sometimes insecure. Most existing systems are either too complicated or not designed according to Freelance needs. Many users also face issues in monitoring budgets, keeping transaction history and identifying suspicious activities. There was a need for a single platform where Freelancers could :-

- 1) manage wallet operations
- 2) track expenses,
- 3) maintain budgets,
- 4) transfer money securely
- 5) monitor financial records properly,

So GigCredit was developed to solve these problems through a secure and easy web platform.

Objectives

The main objectives of this project are:

- To create a secure authentication system using JWT

- 2) To develop a wallet Management system for users
- 3) To allow deposits, with drawals and money transfers.
- 4) To maintain transaction history for users.
- 5) To provide expense tracking and monthly budgeting.
- 6) To generate reports and dashboard summaries.
- 7) To detect suspicious transaction activities.
- 8) To provide admin controls for monitoring the system.
- 9) To build a responsive and user-friendly interface,

Target Users

Primary Users

The main users of GigCredit are freelancers who need a platform to manage wallet activities, expenses and financial records digitally.

These users may include:

- 1) graphic designers
- 2) developers.
- 3) Content writers.
- 4) Video editors.
- 5) online service providers

Secondary Users

The second type of users are administrators:
Admin manage:

- 1) users
- 2) categories
- 3) Transactions
- 4) reports
- 5) suspicious activities

They also monitor system activity and help maintain security and control within the platform.

Functional Requirements

GigCredit was designed to help freelancers manage financial activities through a single platform. The system allows users to create accounts, access dashboards and manage wallet operations securely. Main functions included in the system are:

- - Wallet management
- - deposits and withdrawals
- - money transfers
- - expense tracking
- - monthly budgets
- - transaction history
- - reports and charts
- - suspicious transaction checking
- - admin monitoring panel

Non Functional Requirements

The system was developed with focus on security and smooth navigation. JWT authentication and bcrypt hashing were used to secure user accounts and passwords.

Frontend pages were designed to remain simple and responsive on different screen sizes. Backend validations were added before processing wallet operations and transfers.

The project structure was also divided in separate folders so future updates and maintenance can be handled more easily.

Main NonFunctional aspects considered were:

- 1) security
- 2) responsiveness
- 3) scalability
- 4) usability
- 5) reliability

Technologies used:-

• - GigCredit was developed using MERN stack technologies. React.js and Vite were used on frontend side for user interface and dashboard pages.

Node.js and Express.js were used on backend side for APIs and business logic handling. MongoDB Atlas was used as Cloud database storage.

Other tools and libraries used in the project include:

- - JWT authentication
- - bcrypt hashing
- - Axios
- - Recharts

(4)

User Roles :-

GigCredit contains two main roles inside the system.

User Role :-

- - Normal users can
- - manage wallet activities
- - track expenses
- - create budgets
- - transfer money
- - Request advances
- - view reports and transactions

Admin Role

Admin users monitor overall system activity and manage :

- - users
- - transactions
- - categories
- - suspicious activities

1. HOME PAGE

GigCredit

[Home](#) [Features](#) [About](#) [Contact](#) [Login](#) [Register](#)



Smart Fintech Platform
for Freelancers

Get Started

Why Choose GigCredit?

Wallet Management

Secure Transfers

Expense Tracking

Reports & Analytics

© 2025 GigCredit. All rights reserved.

2. LOGIN PAGE

Welcome Back!

☐ Remember Me [Forgot Password?](#)

Login

Don't have an account? [Register](#)

3. REGISTER PAGE

Create Account

Register

Already have an account? [Login](#)

4. USER DASHBOARD

GigCredit

Dashboard

Wallet

Transfer

Expenses

Budget

Reports

Profile

Logout

User

Wallet Balance
PKR 45,680

Total Income
PKR 120,000

Total Expense
PKR 74,320

This Month Transactions
28

Spending Overview



☒ Food
☐ Transport
☐ Shopping
☐ Others

Recent Transactions

Deposit from Ameer

+ PKR 1,000

12 May 2025

Transfer to Mubashir

- PKR 2,500

11 May 2025

Groceries

- PKR 1,800

08 May 2025

[View All](#)

5. WALLET PAGE

Dashboard

Wallet

Transfer

Expenses

Budget

Reports

Profile

Logout

My Wallet

Current Balance
PKR 45,680



Quick Actions

Deposit

Withdraw

Transfer

Recent Transactions

Deposit

+ PKR 1,000

12 May 2025

Withdraw

- PKR 1,000

12 May 2025

Transfer

- PKR 2,500

11 May 2025

[View All](#)

6. EXPENSE & BUDGET PAGE

Expenses

Budget

Add Expense

Amount

Category

Food

Date

18/05/2025

Description

Add Expense

Monthly Budget

Category

Food

Budget Limit

10,000

Spent

6,200

62%

 Dashboard Users Transactions Wallets Suspicious Activity Reports Settings Logout

Total Users

 128

Total Transactions

 342

Total Balance

 PKR 250,000

Suspicious Activities

 12

Recent Transactions

User	Type	Amount	Status
Ali Khan	Deposit	PKR 5,000	Success
Sara Ahmed	Transfer	PKR 2,000	Success
Bilal Khan	Withdraw	PKR 1,500	Pending
Hina Malik	Deposit	PKR 3,000	Success

[View all transactions →](#)

Recent Suspicious Activities

- Large transaction by Ali Khan 2h ago
- Multiple failed logins by user 5h ago
- Suspicious transfer by Sara Ahmed 1d ago

[View all activities →](#)

Frontend

Frontend of GigCredit was developed using React.js and Vite. Different pages were created for wallet management, expenses, budgets, reports and admin monitoring.

Dashboard-based layout was used so users can quickly move between features without opening too many pages. Cards, charts and recent transaction sections were added to make financial activity easier to understand.

Protected routes were used for pages that require login access. Axios was used for connecting frontend with backend APIs.

The interface was kept simple and responsive so it can work properly on different screen sizes.

Backend structure

Routes folder contain API end points of the system while controllers handle the main request logic. Models folder contains MongoDB schemas for:

- - User
- - Wallet
- - Transaction
- - Expense
- - Budget
- - Category
- - Notification
- - Audit log.

Middle wares were added for authentication, role checking, validation and error handling.

Utility functions and validation files were also separated to improve code management and readability.

API-ROUTES

GigCredit uses REST APIs for frontend and backend communication.

Main route files include:

- - authRoutes.js
- - walletRoutes.js
- - transactionRoutes.js
- - expenseRoutes.js
- - budgetRoutes.js

- - Category Routes.js
- - Notification Routes.js
- - Report Routes.js
- - admin Routes.js
- - User Routes.js.

These routes handle authentication, wallet management transactions, reports and admin operations.

MIDDLEWARES

-> Different middlewares were added in backend for request checking and route protection

auth.js

Used for JWT authentication and protected routes.

role.js

Checks whether user is admin or normal user.

ValidateBody.js

Validates request data before processing APIs.

ValidateObjectId.js

Checks valid MongoDB object IDs.

requireActive.js

Prevents blocked users from accessing protected features.

rateLimiter.js

Limits repeated requests on important routes.

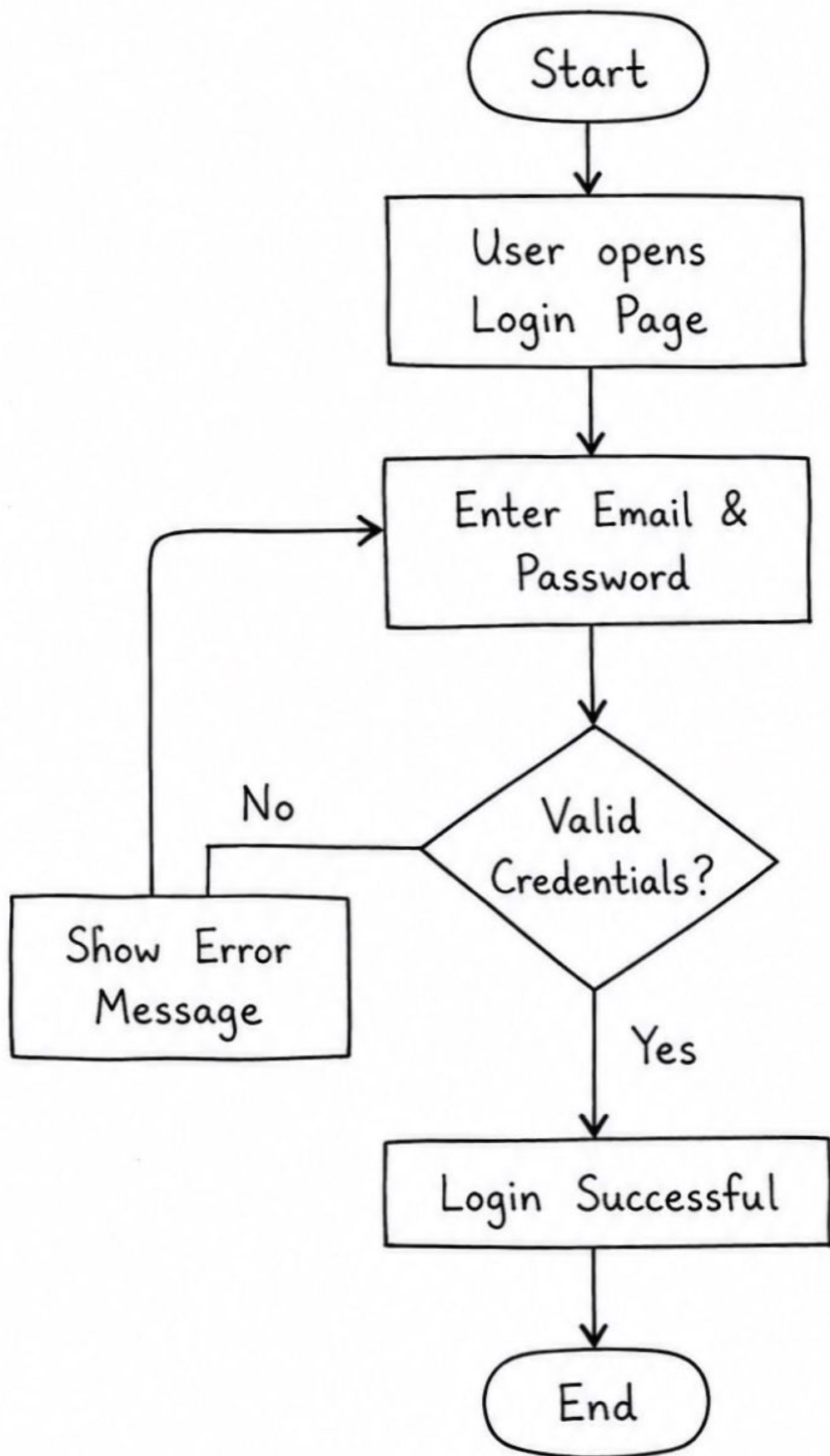
errorHandler.js

Handles backend errors and server response

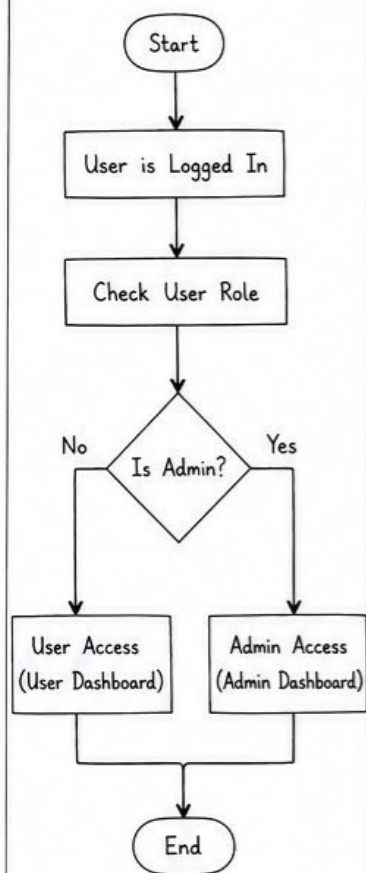
notFound.js

Handles invalid routes and missing endpoints

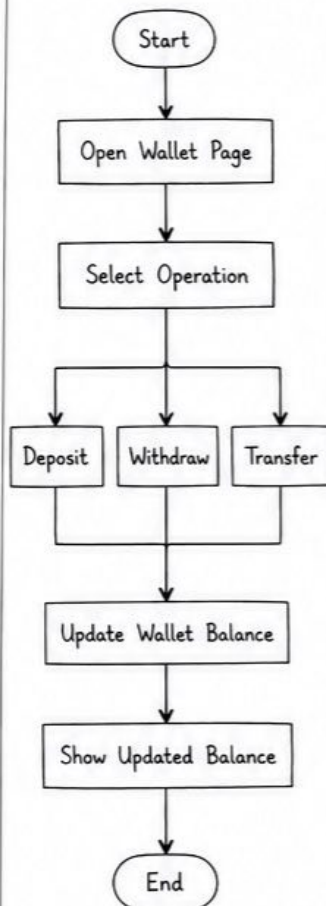
Authentication Flow



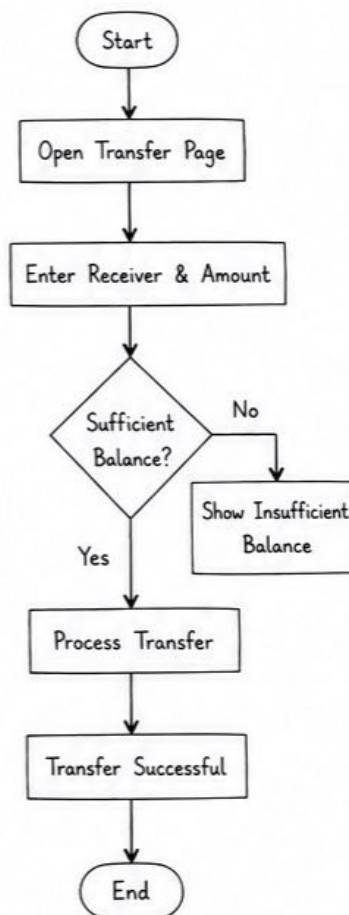
Authorization Flow



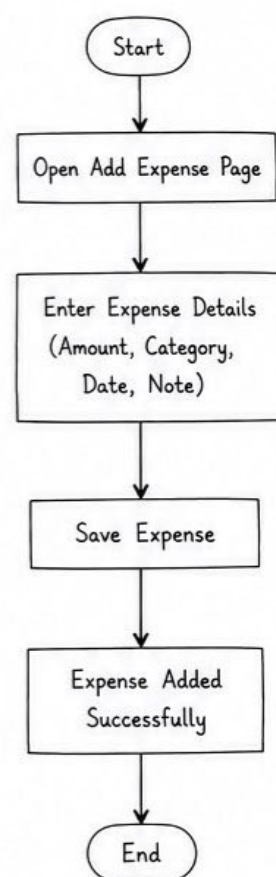
Wallet Flow



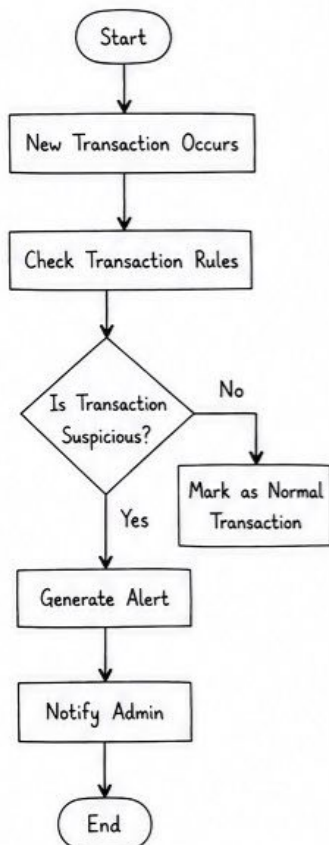
Transfer Flow



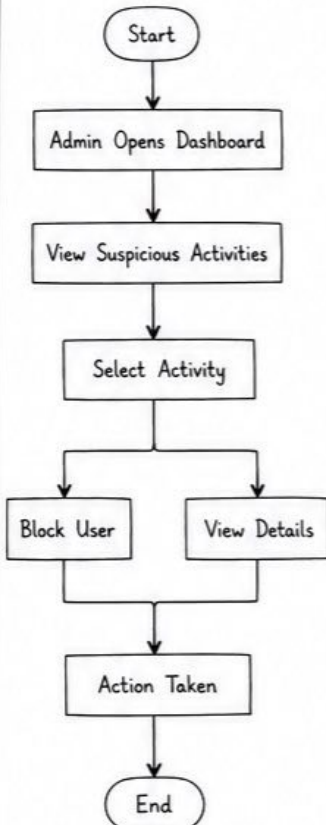
Add Expense Flow



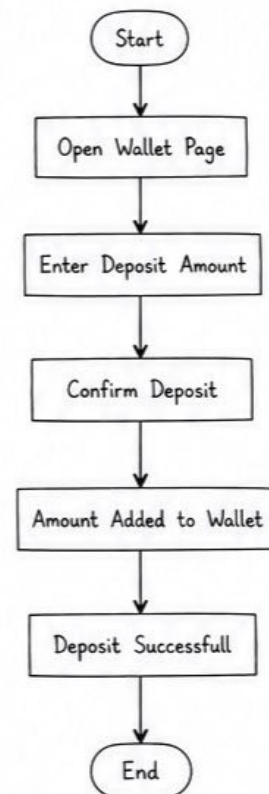
Suspicious Transaction Flow



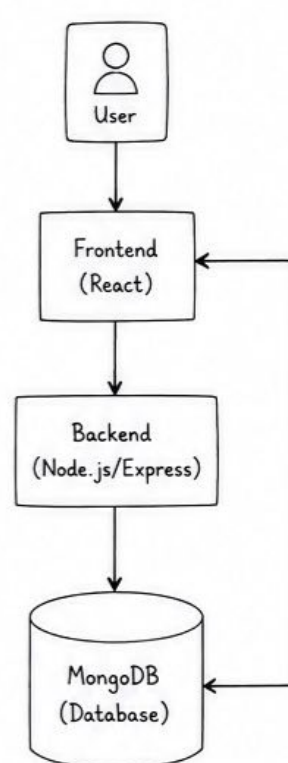
Admin Monitoring Flow



Deposit Flow



Overall System Flow



DATABASE OVERVIEW

Database Design

MongoDB Atlas was used as the cloud database for GigCredit. Mongoose models were created for storing users, wallets, transactions, expenses, budgets and other system records.

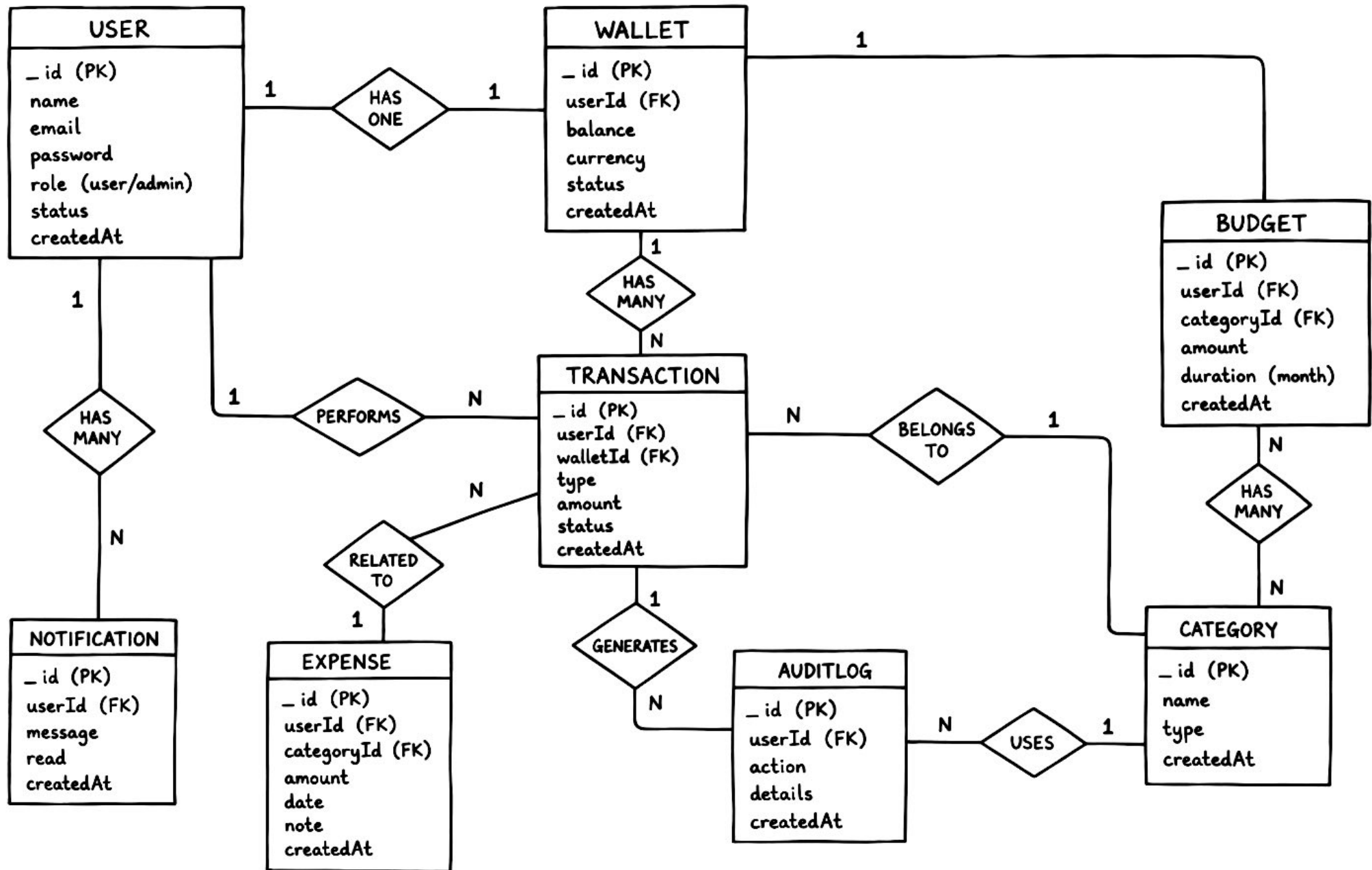
Different collections were connected using ObjectID references so related data can be managed more easily between users and financial activities.

Main collections used in the system are:

- - User
- - Wallet
- - Transaction
- - Expense
- - Budget
- - Category
- - Notification
- - Audit log.

Database validations were also added to maintain proper data handling and avoid invalid records.

ER DIAGRAM (GigCredit)



USER COLLECTION

User collection stores account information and login details of system users.

Main Fields include:

- name
- email
- - password
- - role
- - status
- - createdAt

JWT authentication is connected with user accounts and role-based access is handled using user role information.

Wallet Collection

Wallet collection stores financial balance and wallet-related information for users.

Each user has one wallet connected through ObjectId reference.

Main Wallet fields include:

- - userId
- - balance
- - currency
- - wallet status
- - deposits
- - withdrawals

Wallet updates are handled from backend side after transaction validation.

Transaction Collection

Transaction collection stores records of deposits, withdrawals and transfers.

Main fields include:

- - sender Id
- - receiver Id
- - amount
- - transaction type
- - status
- - Created At

Suspicious transaction flags are also connected with transaction records for admin monitoring.

Expense and Budget collections

Expense collection store user spending records and category information. Budget collection stores monthly budget limits and spending summaries.

Expense fields include:

- - title
- - amount
- - category
- - payment method

Budget fields include:

- - monthly limit
- - category limits.
- - status.
- - warning threshold.

These collections help users track spending and compare expenses against budget limits.

Category and Notification Collections

Category collection stores predefined categories used in expenses and transactions. Notification collection stores alerts and system

Which fields include:
messages related to transactions, budgets and supporting activities.

Notifications help users stay updated about account activity and warnings.

AUDIT LOG COLLECTION

Audit log Collection stores important admin activities and monitoring records.

It helps keep track of actions such as;

- - blocking users
- - Category updates
- - admin activities
- - system changes.

This improves monitoring and accountability inside the platform.

Sample document

```
{  
  name: "Ali",  
  email: "ali@test.com",  
  role: "user",  
  status: "active"  
}
```

AND one wallet example:

```
{  
  userId: "objectId",  
  balance: 50000,  
  currency: "PKR"  
}
```


Security Flow/Measures

Security was an important part of GigCredit because the system handles financial records and wallet operations.

JWT authentication was used for protected routes so unauthorized users cannot access private data.

Passwords were stored using bcrypt hashing instead of plain text storage.

Backend validations were added before deposits, withdrawals and transfers to avoid invalid transactions.

Role-based authorization was also used for admin routes.

Rate limiting middleware was added on sensitive APIs to reduce repeated requests and misuse attempts. Object ID validation was also used before database operations.

Suspicious transactions rules were included to identify unusual wallet activity and improve monitoring from admin side.

Environment variables were used for storing sensitive values such as database URLs and JWT secrets.

Testing Overview

Different modules of GigCredit were tested during development to check API behavior, validations and frontend interaction.

Testing was mainly focused on:

- - authentication
- - wallet operations

- - Transfers
- - budgets and expenses
- - admin controls
- - suspicious transaction

Both successful and invalid cases were checked to verify system behavior and responses

Testing Table 1 - Authentication

Test Case	Expected Results	Status
Valid login	User logged in successfully	Pass
Wrong password	Error message displayed	Pass
Empty fields	Validation message shown	Pass

Testing Table 2 - Wallet

Test Case	Expected Result	Status
Deposit Amount	Balance Updated	pass
Withdraw Valid Amount	Withdrawal Successful	Pass
Withdraw Large Amount	Insufficient balance error	Pass

Testing Table 3 - Transfer

Test Case	Expected Result	Status
Valid Transfer	Transaction Completed	Pass
Invalid Receiver	Error displayed	Pass
Low Balance	Transfer rejected	Pass

Testing Table - 4 - Expenses and Budget

Test Case	Expected Result	Status
Add Expense	Expenses stored	Pass
Create Budget	Budget added	Pass
Exceed Budget	Warning Displayed	Pass

Testing Table 5 - Admin Feature

Test Case	User access restored	Status
Block user	User access Restricted	Pass
View Transaction	Record displayed	Pass
Suspicious Activity	Transaction flagged	Pass

Deployment

Frontend and backend of GigCredit were deployed using cloud hosting platforms while MongoDB Atlas was used for database storage.

Environment variables were configured for secure database connection and JWT authentication.

Deployment helped make the system accessible outside local host for testing and demonstration.

Screenshots -

Limitations

Current version of GigCredit is mainly focused on educational and demo purposes. Real payment gateway integration and live freelance platform APIs were not fully implemented.

Some advanced financial verification and fraud detection features can still be improved in future versions.

Future Work

Future improvements may include:

- real payment gateway integration
- live Upwork/Fiver API connections
- AI Based fraud detection
- mobile Application version
- advanced analytics and reporting

Conclusion

GigGeddit was developed to help freelancers manage wallet activities, expenses and financial records through a single platform. The project also helped understanding of MERN stack development, API integration, authentication and database management.

Gig Credit

Sign in to your account

Welcome back to the place where you left off.

Email

you@example.com

Password

At least 6 characters

[Forgot your password?](#) [Create an account](#)

GIGCREDIT

"A clean ledger is the start of a calm mind."

Every transaction is signed, hashed, and recorded. Suspicious activity is flagged for you, never silently absorbed.



Wallet

Transactions

Expenses

Budgets

Reports

Notifications

Profile

GOOD AFTERNOON

Welcome back, Muskan.

Your GigCredit dashboard — wallet, recent activity, a productive gig bill, at a glance.

WALLET BALANCE

Rs 0

[View →](#)

TRANSACTIONS

0

[View →](#)

CURRENCY

PKR

STATUS

Active ✓

Advance eligibility

new

Enter your pending upwork or Fiverr earnings to see your eligible advance amount (up to 75%) and fee breakdown.

Add earnings

[View reports](#)

Freelance profile

link missing

Keep your profile fresh and up-to-date to get noticed by your ideal clients and increase your earnings.

[Update profile](#)

Monthly outflow

last 0 months

+

No expense activity yet

Record an expense to see your monthly trend.

[Add expense](#)

Quick actions



Withdraw

→

Send to a user

→

Log an expense

→

Manage budgets

→

Recent activity

[View all →](#)

+

No transactions yet

Make your first deposit to get started.

[Add transaction](#)

- Dashboard
- Wallet
- Transactions
- Expenses
- Budgets
- Reports
- Notifications
- Profile

YOUR WALLET

Wallet

Demo funds for testing transfers and expenses. No real money is processed.

Withdraw

Transfer



AVAILABLE BALANCE



Currency: PKR

Lifetime totals

DEPOSITS

Rs 0

RECEIVED

Rs 0

WITHDRAWALS

Rs 0

SENT

Rs 0

Recent transactions

View all →

No activity yet

Make your first deposit to start using your wallet.



- Dashboard
- Wallet
- Transactions
- Expenses
- Budgets
- Reports
- Notifications
- Profile

PERSONAL FINANCE

Expenses

Track your spending. Categorized expenses feed into your budgets and monthly reports.

0

Category

any

Search

Title...

From

dd----yyyy

To

25-Dec-2022

0 items shown · total Rs 0

Reset

By category

Nothing to chart

Log expenses to see a breakdown.

No expenses yet

Add your first expense to start tracking.



- Wallet
- Transactions
- Expenses
- Budgets
- Reports
- Notifications
- Profile

GOOD AFTERNOON

Welcome back, Muskan.

Your GigCredit dashboard — wallet, recent activity, and advance eligibility at a glance.

WALLET BALANCE

Rs 85,000

View —

TRANSACTIONS

1

View —

CURRENCY

PKR

ACCOUNT STATUS

Active ✓

Advance eligibility

new

Enter your pending Upwork or Fiverr earnings to see your eligible advance amount (up to 75%) and fee breakdown.

View reports

Freelance profile

risk scoring

Keep your profile information and payment type updated so your credit analysis stays accurate.

Update profile

Monthly outflow

last 6 months

-

No expense activity yet

Record an expense to see your monthly trend.

Quick actions

- Withdraw —
- Send to a user —
- Log an expense —
- Manage budgets —

Recent activity

View all —

TYPE	DESCRIPTION	DATE	STATUS	AMOUNT
deposit	Initial demo deposit	09 May 2026 00:23	successful	Rs 85,000

▼ gigacredit

📁 auditlogs

📁 budgets

📁 categories

📁 expenses

📁 notifications

📁 transactions

📁 users

📁 wallets

...

```
_id: ObjectId('69fe3817b069bf771603ce27')
transactionId: "TXN-20260508-5SQMYC"
receiverId: ObjectId('69fe3817b069bf771603ce23')
amount: 42000
type: "deposit"
status: "successful"
category: "general"
description: "Initial demo deposit"
suspiciousFlag: false
suspiciousReasons: Array (empty)
createdAt: 2026-05-08T19:23:03.961+00:00
updatedAt: 2026-05-08T19:23:03.961+00:00
__v: 0
```


▼ localhost:27017

▶ admin

▶ blogApp

▶ config

▼ gigcredit

auditlogs

budgets

categories

expenses

notifications

transactions

users

wallets

▶ investmentDB

▶ local

...

```
role : "user"
status : "active"
freelancePlatform : "other"
employerName : "Fiverr (Freelance)"
employmentType : "contract"
monthlyIncome : 150000
createdAt : 2026-05-08T19:23:03.957+00:00
updatedAt : 2026-05-08T19:23:03.957+00:00
__v : 0
```

```
_id: ObjectId('69fe3817b069bf771603ce2b')
name : "Anaya Noor"
email : "anaya@gigcredit.test"
passwordHash : "$2a$10$KuT17kYHvope6JR2.GPd7eEBg9d7X4PqH6zaeJUn8xh1l/8JuOrWm"
role : "user"
status : "active"
freelancePlatform : "other"
employerName : "Freelance / Other"
employmentType : "self_employed"
monthlyIncome : 75000
createdAt : 2026-05-08T19:23:03.965+00:00
updatedAt : 2026-05-08T19:23:03.965+00:00
__v : 0
```

 <http://localhost:5000/api/users/profile>  Save  Share

GET <http://localhost:5000/api/users/profile>



 Docs Params Authorization Headers 7 Body Scripts Settings

Cookies



This request does not have a body

Body 

200 OK 8 ms 1.25 KB  

 JSON  Preview  Visualize



```
1 {
2   "success": true,
3   "message": "Success",
4   "data": {
5     "user": {
6       "_id": "69fe3817b069bf771603ce17",
7       "name": "GigCredit Admin",
8       "email": "admin@gigcredit.test",
9       "role": "admin",
10      "status": "active",
11      "freelancePlatform": "other",
12      "employmentType": "self_employed",
13      "createdAt": "2026-05-08T19:23:03.829Z",
14      "updatedAt": "2026-05-25T12:51:21.086Z",
15      "lastLogin": "2026-05-25T12:51:21.085Z"
16    }
17  }
18 }
```

 **http://localhost:5000/api/users/profile**

 Save  Share 

GET 

http://localhost:5000/api/users/profile

Send 

 Docs

Params

Authorization

Headers 7

Body

Scripts

Settings

Cookies

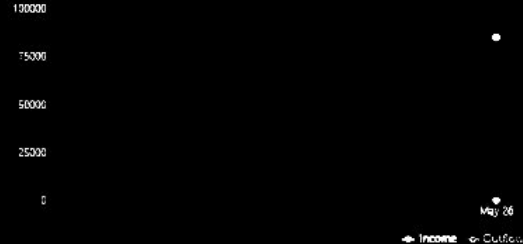
Headers  6 hidden

	Key	Value	Descri...	Bulk Edit	Presets 	
	Authorization	Bearer ey.JhbGciOiJIUzI1NiIsInR5cCI6Ikpz...				
	Key	Value	Description			

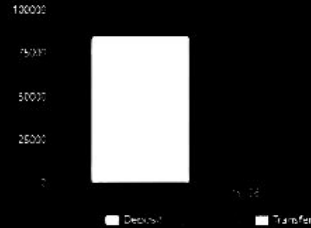
Reports

Track your income, outflow, and type of single as freelance advance eligibility.

Income vs Outflow



Transactions by type



Budget usage

No budgets set
Create a budget on the Budgets page.

Advance eligibility calculator

Enter your pending earnings to estimate your advance up to 75% minus a flat fee

PENDING EARNINGS (PKR)

60000

FEE RATE (%)

4

Total amount pending on your advance will be

Service fee charged on the advance amount (default 4%)